

Matthew Preisser

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EDUCATION

University of Texas at Austin

Aug. 2018 – May 2025

- PhD, Civil Engineering: March 2025 Defense, May 2025 Graduation
- *Dissertation: Description, prediction, & prescription of compounding hazards on society*
- Master's Degree, Environmental and Water Resource Engineering: Dec. 2021
- Master's Degree, Public Affairs: Dec. 2021
- *Thesis: Intersecting near real-time fluvial and pluvial inundation estimates with sociodemographic vulnerability to quantify a household flood impact index*

Auburn University

Aug. 2014 – May 2018

- Bachelor's Degree, Biosystems Engineering, minors in Sustainability Studies & German: May 2018
- *Thesis: Anaerobic digestion of food waste and poultry litter for biogas production*
- Alabama Society of Professional Engineers Outstanding Student of the Year (2018)
- Auburn University Biosystems Engineering Outstanding Student of the Year (2018)

PROFESSIONAL APPOINTMENTS

Postdoctoral Researcher

Aug. 2025 - Present

Department of Civil, Environmental and Geomatic Engineering

ETH Zurich

- Advisor: Dr. Paola Passalacqua

Graduate Research Assistant

Aug. 2018 – May 2025

Maseeh Dept. of Civil, Architectural and Environmental Engineering

University of Texas at Austin

- Advisor: Dr. Paola Passalacqua
- Committee: Dr. R. Patrick Bixler, Dr. Matt Bartos, & Dr. Stephen Boyles
- Primary Funding: NSF GRFP, NASA FINESST
- Research Collaborations, DoE SETx-IUFL, PlanetTexas2050, & NASA ATX Climate Atlas

GIS Research Analyst

Jun. 2018 – Aug. 2018

NASA DEVELOP

Marshall Space Flight Center

- Advisors: Dr. Jeffrey Luvall & Dr. Robert Griffin
- *Assessing Water Quality in Thailand's Chao Phraya Watershed by Modeling Sediment Conc.*
- Primary Funding: NASA DEVELOP and SSAI

Undergraduate Research Fellow

Jan. 2017 – May 2018

Department of Biosystems Engineering

Auburn University

- Advisor: Dr. Brendan Higgins
- *Waste products for algae and biogas and sustainable energy production*
- Primary Funding: Auburn University Honors College

UNDER REVIEW AND IN PREPARATION PUBLICATIONS

1. **Preisser, M.** and Passalacqua, P. (2026). Socio-Hydrology Modeling Captures How Inequalities Impact Community Flood Resilience. *Water Resources Research* (Accepted).

PEER REVIEWED PUBLICATIONS

7. Rosenheim, N., Mezei, L., **Preisner, M.**, Brelsford, C., Bixler, R. P., Meyer, F., and Passalacqua, P. (2026). Where social vulnerability indices diverge: Navigating the decision cascade for local site selection. *Natural Hazards* 122.11, p. 476. DOI: 10.1007/s11069-026-08249-9.
6. **Preisner, M.** and Passalacqua, P. (2025). Remote Sensing Improves Multi-Hazard Flooding and Extreme Heat Detection by Fivefold Over Current Estimates. *AGU Advances* 6.2, AV001667. DOI: 10.1029/2025AV001667.
5. Bixler, R. P., Paul, S., Bhakta, D., Farchy, T., Olson, J., **Preisner, M.**, and Passalacqua, P. (2023). Adaptive Governance for Disaster Risk Reduction. *Handbook on Adaptive Governance*. Edward Elgar Publishing, pp. 233–251. ISBN: 978-1-80088-824-1. DOI: 10.4337/9781800888241.00026.
4. **Preisner, M.**, Passalacqua, P., Bixler, R. P., and Boyles, S. (2023). A Network-Based Analysis of Critical Resource Accessibility during Floods. *Frontiers in Water* 5. DOI: 10.3389/frwa.2023.1278205.
3. **Preisner, M.**, Passalacqua, P., Bixler, R. P., and Hofmann, J. (2022). Intersecting Near-Real Time Fluvial and Pluvial Inundation Estimates with Sociodemographic Vulnerability to Quantify a Household Flood Impact Index. *Hydrology and Earth System Sciences* 26.15, pp. 3941–3964. DOI: 10.5194/hess-26-3941-2022.
2. Bixler, R. P., Paul, S., Jones, J., **Preisner, M.**, and Passalacqua, P. (2021). Unpacking Adaptive Capacity to Flooding in Urban Environments: Social Capital, Social Vulnerability, and Risk Perception. *Frontiers in Water* 3. DOI: 10.3389/frwa.2021.728730.
1. Chaump, K., **Preisner, M.**, Shanmugam, S. R., Prasad, R., Adhikari, S., and Higgins, B. (2019). Leaching and Anaerobic Digestion of Poultry Litter for Biogas Production and Nutrient Transformation. *Waste Management* 84, pp. 413–422. DOI: 10.1016/j.wasman.2018.11.024.

SOFTWARE PACKAGES

SVInsight <i>Open-source tool for estimating social vulnerability indices</i>	Github: v1.1.2
pygeoflood <i>Near-real-time tool for estimating fluvial, pluvial, and coastal flooding</i>	Github: Beta
pySocioHydro <i>Socio-hydrology dynamical systems modeling package</i>	Github: Beta

GRANTS AND FELLOWSHIPS

Total Grants and Fellowships Received (2018-Present): \$274,750

7. Consortium of Universities for the Advancement of Hydrologic Science Hydroinformatics Fellow
 - *Building out a near-real-time compound inundation python package*
 - Team: Dr. Paola Passalacqua and Mark Wang
 - Obligated Funding: \$5,000 total, \$2,500 to Preisner
 - Period: April 2024 – Feb. 2025
6. NASA Future Investigators in NASA Earth and Space Science and Technology (FINESST) Fellow
 - *Application of Earth Observation Data in Multi-Layer Network Analysis of Flood Hazards, Social Vulnerability, and Risk*
 - Advisor: Dr. Paola Passalacqua
 - Obligated Funding: \$100,000

- Period: Aug. 2022 – Sept. 2024
5. National Science Foundation Graduate Research Internship Program (GRIP) Fellow
 - *USGS National Land Imaging Program Public Affairs and Science Communication Internship*
 - Mentor: Timothy Stryker
 - Obligated Funding: \$5,000
 - Period: Sept. 2019 – Aug. 2022
 4. National Science Foundation Graduate Research Fellow (NSF-GRFP)
 - *Incorporating flood mapping with social disparities to identify at risk communities during extreme weather events*
 - Advisor: Dr. Paola Passalacqua
 - Obligated Funding: \$150,000
 - Period: May 2020 – Aug. 2020
 3. Strauss Center's Brumley Next Generation Fellowship
 - *Visualizing climate change threats in relationship to power generation capabilities in India*
 - Mentor: Dr. Joshua Busby
 - Obligated Funding: \$5,000
 - Period: Sept. 2019 – May 2020
 2. Supplemental Professional Development and Travel Grants
 - UT Civil, Architectural, and Environmental Engineering Kolodzey Travel Grant: \$1,000
 - UT Division of Diversity and Engagement Professional Development Award (2023): \$250
 - UT Dean's Prestigious Supplement Fellowship (2019-2023): \$11,000
 1. Pre-2018 Academic Scholarships
 - Birdsong Endowed Engineering Scholarship (2017)
 - Fred A. Kummer Endowed Scholarship (2015)
 - Hop King Endowed Scholarship (2015)
 - Auburn University Presidential Academic Scholarship (2014-2018)

HONORS AND AWARDS

- Civil Engineering Symposium 2nd Place Poster (2021)
- Patagonia Sustainability Case Competition International Top 10 Finalist (2019)
- Auburn University Biosystems Engineering Outstanding Student of the Year (2018)
- Alabama Society of Professional Engineers Outstanding Student of the Year (2018)
- Inducted into Phi Beta Kappa Society (2018)
- Mark A. Spencer Creative Mentorship Award (2018)
- Semi-Finalist for the Fulbright Scholarship (2018)
- ReNUWIt Best Overall Research Poster (2017)
- Nominated for the Truman and Udall Scholarships (2017)
- Inducted into Tau Beta Pi Honor Society (2016)

PROFESSIONAL SERVICE AND VOLUNTEERING

Journal Reviews

Communications Earth & Environment; Frontiers in Water; Geomatics, Natural Hazards, & Risk; Hydrological Processes; International Journal of Urban Sustainable Development; International Journal of Digital Earth; International Journal of Disaster Risk Reduction; Journal of Infrastructure Systems; npj Natural Hazards; Risk Analysis.

AGU Outstanding Presenters Advisory Group

Mar. 2026 – Present

Representative, advisory group for student & early career presenter award program

AGU Council

Jan. 2023 – Present

Elected representative, advising leadership committee on scientific affairs

AGU H3S Committee

Jan. 2022 – Dec. 2024

Treasurer of Hydrology Section Student Subcommittee

AGU Thriving Earth Exchange Community Science Fellowship

Mar. 2021 – Mar. 2022

Coordinated a team of volunteers to address flood and lead related issues

NASA DEVELOP Ambassador

Aug. 2018 – May 2019

Program ambassador corps representative

MENTORING EXPERIENCE

Graduate Students

- Alessandro Derighetti, Master's Thesis, Civil and Environmental Engineering, 2026
 - * *Understanding the role of spatial rainfall patterns on streamflow generation in the Swiss Alps*
 - * Co-advisor: Dr. Harsh Beria

Undergraduate Students

- Thomas Rüegg, Bachelors of Civil Engineering, Bachelor's Thesis, ETH Zürich, 2026
- Sahara Tassin, UT Austin Environmental Science Institute REU 2024
- Bailey Hardage, NASA ATX Climate Atlas Program UT Austin, Sept. 2023 – May 2024

INVITED PRESENTATIONS

2. Passalacqua, P., **Preisser, M.**, and Wang, M. "Acute on Chronic Stressors in Communities: When Flooding Is Not the Only Issue" – AGU Annual Meeting, NH23A-2272, Dec. 2024. (Oral).
1. Passalacqua, P., **Preisser, M.**, Wang, M., Bixler, P., Hooks, A., Hofmann, J., Haselbach, L., Moftakhari, H., Evans, H., Thies, C., and Maidment, D. "Preparing for Future Floods: Leveraging Remotely Sensed Data, Modeling, and Social Science Information in a Multilayer Network Approach" – AGU Annual Meeting, H46D-01, Dec. 2022. (Oral).

PRESENTATIONS

17. **Preisser, M.** and Passalacqua, P. "Adaptive Capacity Disparities: Capturing Resiliency Inequalities in Socio-Hydrological Systems" – EGU General Assembly, EGU26-11114, May 2026. (Oral).
16. **Preisser, M.** and Passalacqua, P. "Adaptive Capacity Disparities: Capturing Resiliency Inequalities in Socio-Hydrological Systems" – AGU Annual Meeting, GC079, Dec. 2025. (Poster).
15. **Preisser, M.** and Passalacqua, P. "Remote sensing improves multi-hazard flooding and extreme heat detection by fivefold over current estimates" – AGU Annual Meeting, NH011, Dec. 2025. (Oral).

14. **Preisser, M.** and Passalacqua, P. “The Compounding Impacts of Hydro-Meteorological Hazards on Society” – ZHydro, ETH Zurich, Zurich, Switzerland, Nov. 2025. (Oral).
13. **Preisser, M.** and Passalacqua, P. “Missing Hazards: Improving detection and delineation methods of multi-hazard events and assessing their impact on vulnerable populations” – AGU Annual Meeting, GC31V-0118, Dec. 2024. (Poster).
12. Wang M. **Preisser, M.** and Passalacqua, P. “Near Real-Time Compound Flood Inundation Python Package” – AGU Annual Meeting, NH41D-2342, Dec. 2024. (Poster).
11. **Preisser, M.**, Passalacqua, P., and Bixler, P. “A network-based disaster resilience metric for estimating individuals’ loss of access to critical resources during flooding” – EGU Annual Meeting, Vienna, Austria, Apr. 2023. (Oral).
10. **Preisser, M.**, Passalacqua, P., Bixler, P., Kamath, H., and D., N. “Multiple hazards from multiple satellites: Analyzing exposure to cascading hazards using daily Earth observation data from the past decade” – AGU Annual Meeting, NH33D-0822, Dec. 2023. (Poster).
9. Karimaghaei, S., Passalacqua, P., **Preisser, M.**, and Bixler, P. “Identifying Morphological Hot Spots and Social Vulnerabilities along the Gulf Coast using Remote Sensing” – AGU Annual Meeting, EP15B-1087, Dec. 2022. (Poster).
8. **Preisser, M.**, Passalacqua, P., and Bixler, P. “Helping Those Who Need It Most: Open Access Tool for Modeling Flood Hazards and Community Adaptive Capacity in Near-Real Time” – AGU Annual Meeting, H55M-0745, Dec. 2022. (Poster).
7. **Preisser, M.**, Passalacqua, P., and Bixler, P. “Interconnecting Networks of Social Vulnerability, Resource Access, and High-Resolution Inundation to Quantify Household Flood Impact” – AGU Annual Meeting, H55V-0993, Dec. 2021. (Poster).
6. Newman, T., Wu, Z., Reed, B., Stryker, T., and **Preisser, M.** “Assessing COVID-19 environmental impacts through remote sensing” – AGU Annual Meeting, U012-05, Dec. 2020. (Oral).
5. **Preisser, M.**, Moler, N., Benedict, K., Shapiro, C., Straub, C., and Stryker, T. “Expanding the Societal Benefits of Earth Observations through Leading Practices for Public Private Partnerships” – AGU Annual Meeting, SY003-0002, Dec. 2020. (Poster).
4. **Preisser, M.**, Passalacqua, P., and Bixler, P. “A Multiplex approach to integrate social vulnerability into urban flood mapping” – AGU Annual Meeting, H139-0015A, Dec. 2020. (Poster).
3. Garcia, K., **Preisser, M.**, and Nauman, C. “Assessing Water Quality in Thailand’s Chao Phraya Watershed by Modeling Sediment Concentration and Urban Footprint” – AAG Annual Meeting, Washington DC, Apr. 2019. (Poster).
2. Garcia, K., **Preisser, M.**, and Nauman, C. “Assessing Water Quality in Thailand’s Chao Phraya Watershed by Modeling Sediment Concentration and Urban Footprint” – Wernher von Braun Memorial Symposium, University of Huntsville, Apr. 2019. (Poster).
1. **Preisser, M.** and Higgins, B. “Anaerobic digestion of poultry litter and food waste for biogas production” – National Conference of Undergraduate Research, University of Oklahoma, Apr. 2019. (Poster).

CONVENED SESSIONS AND TOWN HALLS

7. Patterson, E., Akerele, A., Fields, M., Ellis, E., Roy, T., and **Preisser, M.** “Epic Fails and Bumpy Trails: Owning and Learning from the Messy Side of Geoscience”, H12I, 2025.
6. **Preisser, M.** and Miralha, L. “Hydrology Student, Postdoc, and Early Career Flash Talk Geoburst Session” – American Geophysical Union Annual Meeting, H12I, 2025.
5. Miralha, L., Peterson, D., **Preisser, M.**, Akerele, A., Che, Y., and Lerbeck, J. “Intersections in Science Roundtable: Building Cross-Disciplinary Connections for Early Career Researchers” – American Geophysical Union Annual Meeting, AGU Learning Session, 2024.
4. Ellis, E., Becker, P., Price, A., and **Preisser, M.** “Communicating Science Beyond the Paper: Thinking Outside the Boxplot” – American Geophysical Union Annual Meeting, INV13A, 2023.
3. Peterson, D., Miralha, L., Lerbeck, J., **Preisser, M.**, and Allen, J. “Where and How Can AGU Assist? Discussion About the Future of Early-Career and Student Members Within AGU” – American Geophysical Union Annual Meeting, TH33G, 2023.
2. **Preisser, M.**, Jha, A., Ali, J., Sui, X., and Adelsperger, S. “Building Your Network: Sharpening the Soft Skills of Science” – American Geophysical Union Annual Meeting, TH43B, 2023.
1. Patterson, J., Miralha, L., and **Preisser, M.** “Building Your Network: AGU-H3S Networking” – American Geophysical Union Annual Meeting, TH13D, 2022.

ADDITIONAL WORK EXPERIENCE

GIS Analyst Intern <i>HDR Engineering</i>	Mar. 2019 – Aug. 2019 <i>Austin, Texas</i>
Sustainability Senior Student Associate <i>University of Texas at Austin Office of Sustainability</i>	Aug. 2018 – May 2019 <i>Austin, Texas</i>
ReNUWIT REU Program <i>Colorado School of Mines: Advised by Dr. Tzahi Cath</i>	May 2017 – July 2017 <i>Golden, Colorado</i>
DoE SULI Intern <i>Energy & Environment. Directorate, Pacific Northwest National Lab</i>	May 2016 – July 2016 <i>Seattle, Washington</i>
Engagement and Outreach Intern <i>Auburn University Office of Sustainability</i>	Aug. 2015 – May 2018 <i>Auburn, Alabama</i>